Significant figures

1 State the number of significant figures in the following numbers:

a 108 486
b 5105
c 0.20
d 10.20
e 84.00
f 0.007 193 0
g 0.004 58
h 2.0058

2 State the smallest possible number of significant figures in the following numbers:

a 1 500 000
b 26 500
c 1 060 000
d 82 420 000

3 A rainfall gauge has markings every 10 mL. If 369 mL of rain fell in this gauge, to how many significant figures would the rain be recorded?

4 Beth just bought a house for $858 400. When asked for the price by her friend, she said she paid around $860 000. To how many significant figures did Beth round the price in her answer?

5 A number has been rounded to two significant figures. If the rounded number is 4500, find the largest possible value of the original number.

6 Round the following numbers to two significant figures:

a 2584
b 10 694
c 9249
d 215.99
e 0.7218
f 0.0297
g 4.95
h 0.006 037 736

7 Round the following numbers to three significant figures:

a 1771
b 461 585
c 6484
d 20 994
e 9248
f 539.99
g 0.5628
h 0.073 582

8 Round the following numbers to four significant figures:

a 801 600 713
b 0.060 070 047
c 11.8483
d 2.159 76
e 4.504 13
f 1.599 64
g 90.0335
h 0.598 71

9 Round 7231.0108 to five significant figures.

10 Consider the number 1083.

a Round 1083 to:

i One significant figure

ii Two significant figures

b Find the difference between these two rounded values.

11 A number, when rounded to one significant figure, is 3000.

a Find the largest possible value of the original number.

b Find the smallest possible value of the original number.

c Find the difference between the largest and smallest values.

12 A number, when rounded to one significant figure, is 500. The original number is a multiple of 4, and halving it gives you a square number. Find the original number.

Absolute error

13 Which of the following measurements is most precise?

A 2959 mL **B** 920.0 mL **C** 2.95 L

14 In the following calculation, which value is the least precise?

$$149.4 \text{ cm} + 7.6 \text{ cm} - 3.20 \text{ cm}$$

15 State the absolute error of the following:

a A measurement of 30 kg, which was measured to the nearest kilogram.

b A measurement of 400 g, which was measured to the nearest gram.

c The amount $\$700$, which was calculated to the nearest $\$10$.

16 State the absolute error of each of the following measurements obtained from measuring devices:

a 18.9 mL

b 269 mm

c 9.9 g

d 18 s

e 98.6°F

f 15.12 kg

g 17.8 m

h 117.38°C

17 A chef measures the amount of olive oil to b[redacted]ed in a cup marked in milliliters. Find the absolute error of any measurement she makes.

18 The number of people at a concert is approximately 5050. Explain why we cannot find the absolute error of this approximation.

19 Perform the following calculation, leaving your answer in meters and using the least precise measurement provided:

$$11.19 \text{ m} - 1.185 \text{ m} + 150.81 \text{ cm}$$

20 For each of the following expressions:

i State the number of significant figures given in the least precise measurement.

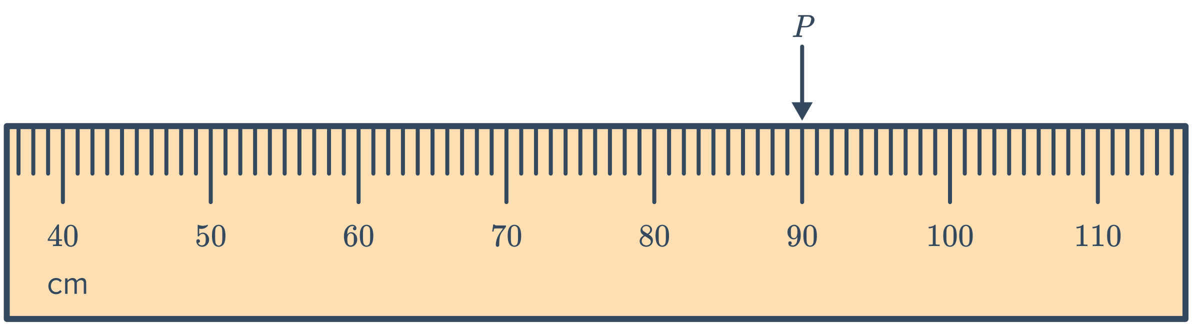
ii Now, calculate the result, rounded to the least number of significant figures.

a $9.17 \text{ m} \times 9.790 \text{ m} \times 4.70 \text{ m}$

b $418 \text{ mL} \times 5.7 \div 934.5$

Limits of accuracy

21 Consider the tape measure below:



a What is the smallest unit marked on the tape?

b What should be recorded for measurement P?

c What is the absolute error of measurement P?

d State whether the following could be the actual measurement of P:

i 89.02 cm

ii 90.64 cm

iii 89 cm

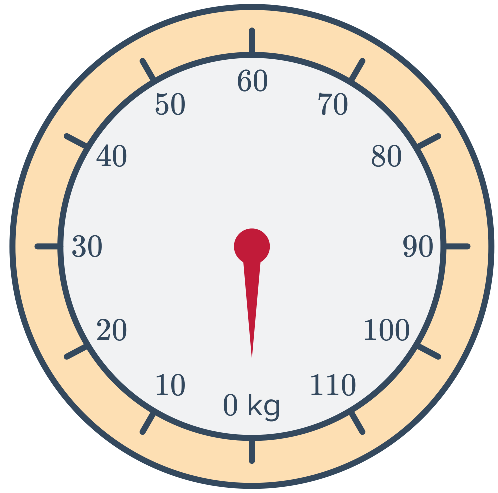
iv 90.03 cm

22 Consider the following scale:

a What is the smallest unit labeled on the scale?

b What is the absolute error?

c If a certain object is measured at 70 kg, what are the lower and upper bounds of this measurement?



23 Consider the following speedometer:

a At what speed is being indicated on the speedometer?

b What is the absolute error of this speed reading?

c What would the maximum speed limit need to be to ensure the driver is not speeding?

24 The length of a piece of rope is measured to be 19.99 m using a ruler. What is the upper bound of the largest possible length of this rope?

25 The height of a tower is measured as 12.1 m. What is the shortest possible height of the tower?

26 For the following measurements, find:

i The upper bound.

ii The lower bound.

a A distance measured to be 13.45 km.

b A distance measured to be 6.4 mi.

c A height measured to be 5 yd.

d A bag of sugar weighs 14 kg to the nearest 10 grams.

e Puncak Jaya, Indonesia’s greatest mountain, is 4884 m high rounded to the nearest meter.

f The cost of a CD lie if it is known to be $\$50$ correct to the nearest $\$5$.

27 State the maximum and minimum possible number of the following:

a A town's population is estimated to be 170 people, to the nearest 10 people.

b A stack of paper is estimated to have 3000 sheets to the nearest 100 sheets.

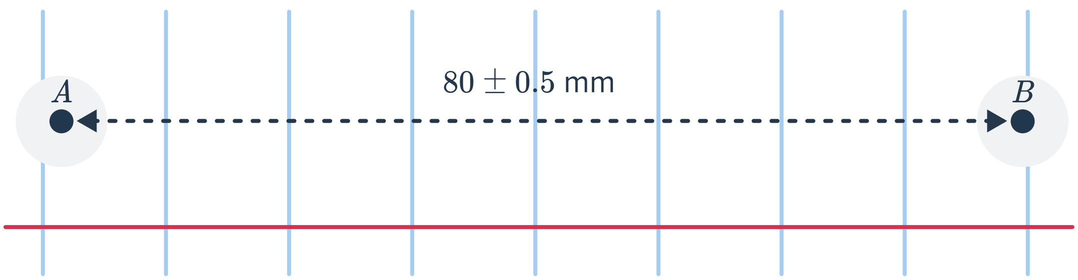
28 $c = 4$ and $d = 8$ have both been rounded to the nearest whole number.

a Calculate the lower bound of the value of $c \times d$.

b Calculate the upper bound of the value of $\dfrac{c}{d}$.

Applications

29 To make paper easier to file, two holes of diameter $6 \pm 0.5 \text{ mm}$ are made using a hole punch. The centers of the two holes are $80 \pm 0.5 \text{ mm}$ apart, as shown in the following diagram:



- **a** What is the smallest unit of measurement of the holes?
- **b** Find the closest possible distance between points A and B.

30 A field has dimensions $15.4 \text{ m} \times 17.6 \text{ m}$, to the nearest 10 cm.

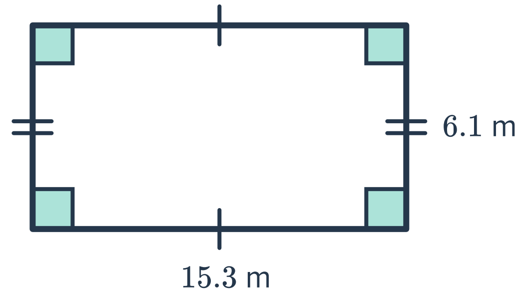
- **a** What are the upper and lower bounds of the area of the field?
- **b** What is the upper and lower bounds of the perimeter of the field?

31 The dimensions of a room are given as 3.6 m by 8.6 m, to the nearest 10 cm. What area of carpet should be purchased to make sure it covers the whole room?

32 A school field has a 100 yd track that is measured to the nearest yard. The record for the fastest 100 yd sprint is 13.9 seconds rounded to the nearest 0.1 second.

- **a** What is the fastest speed, in meters per second, the record holder could have run? Round your answer to two decimal places.
- **b** What is the slowest speed, in meters per second, the record holder could have run? Round your answer to two decimal places.

33 The dimensions of the rectangle have been measured to the nearest 0.1 of a meter.



a Find the upper bound for the maximum possible area of the rectangle.

b What is the minimum possible area of the rectangle?